Assignment 4: Vital Sign Description

Chosen Vital Sign from Emergency Triage Dataset: Mean Arterial Pressure (MAP)

The mean arterial pressure (MAP) is defined as the average force of blood being pushed against the walls of the arteries as it is being pumped through the body, otherwise known as arterial pressure, throughout one cardiac cycle. The mean arterial pressure is influenced by cardiac output and systemic vascular resistance (SVR) which is the force the heart must overcome to pump blood through the systemic circulation. In the triage system, nurses use it as a crucial and rapid indicator for organ perfusion. A patient's body may be good at compensating for early blood loss or infection. As the blood vessels narrow or the heart pumps faster, their systolic pressure may look deceptively normal e.g. 100/50 mmHg. However, if the nurse calculates the MAP from the blood pressure readings, the results are dangerously low. MAP helps nurses catch hidden and early stage shock that a quick systolic pressure reading might miss.

MAP is used to assess the adequacy of blood perfusion to vital organs. The quantitative ranges to determine where MAP is within normal or abnormal range are as follow:

- (<60 mmHG): A MAP below 60 mmHG indicates that vital organs may not receive sufficient blood flow. This can lead to ischemia, brain damage and or organ failure.
- (70 to 100 mmHG): The normal physiological range and indicates adequate blood flow throughout the body.
- (>100-110 mmHG): An elevated MAP value suggests that there is excessive stress on the arteries and heart, which can eventually lead to blood clots or organ damage.

Reference: https://www.ncbi.nlm.nih.gov/books/NBK538226/